

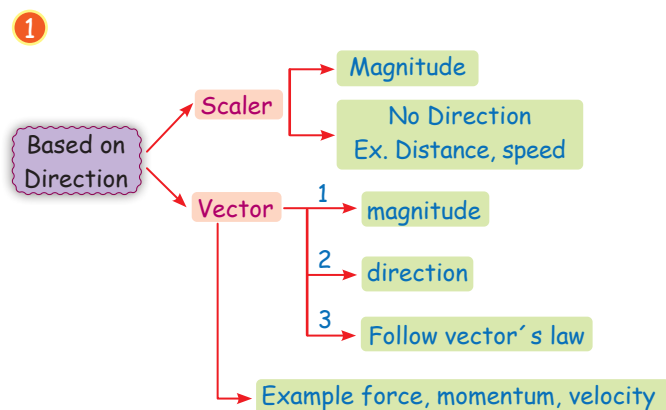
Toh chaliye, hamari is beautiful journey ko start karte hain, pehle chapter Unit and Dimensions se. Ye chapter Advance/Mains/NEET teeno ke liye important hai. Mujhe pata hai, pehle chapter mein apka excitation level bahut high hota hai. From basic to advance ka content hum is chapter mein cover karenge. Main bas itna bolunga, ho sakta hai shuru mein ye chapter apko thoda mushkil lage, but at the end of physics, ye chapter apko sabse aasan lagega. From basic to advance, all relevant theory and questions I will cover in this book. Toh chaliye shuru karte hain, are you ready?



PHYSICAL QUANTITIES

All quantities that can be measured are called physical quantities. e.g. time, length, mass, force, work done, etc.

Classification of Physical Quantity



Kya hota hai vector? Kya hota hai vector's law? Abhi humein in par bilkul dhyaan nahi dena hai. Ye cheezein hum vector waale chapter mein detail mein padhenge.



2 Based on dependency

Fundamental physical quantity

Derived physical quantity

Based on their Dependency

- 1 Fundamental or base quantities:** A set of physical quantities which are completely independent of each other and all other physical quantities can be expressed in terms of these physical quantities is called Set of Fundamental Quantities.
- 2 Derived quantities:** The quantities which can be expressed in terms of the fundamental quantities are known as derived quantities. Ex. Speed (= distance/time), volume, acceleration, force, pressure, etc.



Haan Ye Karlo Pehle

Fundamental or base quantities

S. No.	Physical quantity	SI unit	Symbol
1.	Length	metre	m
2.	Mass	kilogram	kg
3.	Time	second	s
4.	Temperature	Kelvin	K
5.	Electric current	ampere	A
6.	Luminous intensity	candela	cd
7.	Amount of substance	mole	mol

A quantity can be expressed as numerical value and unit.

$$\text{Quantity (Q)} = n_1 U_1 = n_2 U_2$$

Physical quantity to measure.

$n_1 \rightarrow$ numerical value in 1st system of measurement

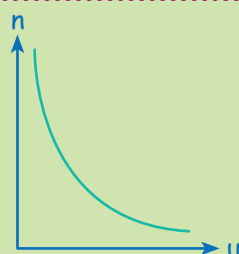
$U_1 \rightarrow$ Units value in 1st system of measurement

$n_2 \rightarrow$ numerical value in 2nd system of measurement

$U_2 \rightarrow$ Units value in 2nd system of measurement

If $U_1 > U_2 \Rightarrow n_1 < n_2$

If unit is bigger then numerical value is smaller and vice versa



Classification of Units

meter kilogram sec. MKS System	CGS System	FPS System
mass $\rightarrow$ kg	mass $\rightarrow$ gram	mass $\rightarrow$ pound
length $\rightarrow$ m	length $\rightarrow$ cm	length $\rightarrow$ foot
time $\rightarrow$ sec	time $\rightarrow$ sec	time $\rightarrow$ sec

Q. Convert Density = 50 kg/m³ into CGS.

$$\text{Sol. Density} = 50 \text{ kg/m}^3 = \frac{50 \times 1000 \text{ gm}}{(100 \text{ cm})^3} = .05 \text{ gm/(cm)}^3$$

Q. Convert Density = 0.6 gm/cc into MKS.

$$\text{Sol. Density} = 0.6 \text{ gm/cc} = \frac{0.6 \times \frac{\text{kg}}{1000}}{\left(\frac{1}{100} \text{ m}\right)^3} = \frac{600 \text{ kg}}{\text{m}^3}$$

DIMENSIONS

Dimensions of a physical quantity are the powers (or exponents) to which the base quantities are raised to represent that quantity.

The physical quantity that is expressed in terms of base quantities is enclosed in square brackets to remind that the equation is among the dimensions and not among the magnitude, such expression for physical quantity is called dimension formula.

meter wala 'm' nahi hai

Mass has dimension [M]

Length has dimension [L]

Time has dimension [T]

Temp. has dimension [K]

Electric current [A]



Q. Write dimensional formula of speed.

$$\text{Sol. Speed} = \frac{\text{Distance}}{\text{Time}}$$

Dimensionally we can write

$$[\text{Speed}] = \left[\frac{\text{Distance}}{\text{Time}} \right] = \left[\frac{L}{T} \right] = [LT^{-1}]$$

So, dimensional formula of speed can be written as $LT^{-1} = [M^0 L^1 T^{-1}]$

Here

Dimension of mass $\rightarrow 0$

Dimension of length $\rightarrow 1$

Dimension of time $\rightarrow -1$



Sir, speed ka Dimensional Formula (DF) kya hai? Kya hum only LT^{-1} likh sakte hain? Kya ise box mein band karna zaroori hai?

Waise toh speed ka DF hum LT^{-1} , $[LT^{-1}]$, $[M^0 LT^{-1}]$ teeno likh sakte hain, lekin box lagana acchi baat hai. Hum aksar physical quantity ko M, L, T mein represent karte hain. Agar tum school mein exam de rahe ho toh box zaroor lagao, taaki school wale master sahab number na kaate. Lekin, agar tum JEE ka test/mock test/question practice kar rahe ho, toh box lagane mein time waste mat karna.



Q. Find the dimensions of mass length & time in density.

Sol. Formula of density is

$$[\text{Density}] = \frac{[\text{mass}]}{[\text{volume}]}$$

$$[\text{Density}] = \frac{M}{L \cdot L \cdot L} = \frac{M}{L^3} \Rightarrow [M^1 L^{-3} T^0]$$

Hence in density we can say

Dimensional of mass $\rightarrow 1$

Dimension of length $\rightarrow -3$

Dimension of time $\rightarrow 0$

1 Fundamental or base units:

The units of fundamental quantities are called *base units*. In SI there are seven base units.

2 Derived units:

The units of derived quantities or the units that can be expressed in terms of the base units are called *derived units*.

Ex. Unit of speed can be derived from units of distance & unit of time.

$$\text{Unit of speed} = \frac{\text{unit of distance}}{\text{unit of time}} = \frac{\text{metre}}{\text{second}} = \text{ms}^{-1}$$

Dimensions of Quantities Related to Mechanics

S.N.	Quantity	Formula	Unit	Dimension
1.	Velocity or speed (v)	$v = \frac{d}{t} = \frac{\text{Displacement or Distance}}{\text{Time}}$	m/s	$[M^0 L^1 T^{-1}]$
2.	Acceleration (a)	$a = \frac{\Delta v}{\Delta t} = \frac{\text{Change in velocity}}{\text{Change in time}}$	m/s ²	$[M^0 L T^{-2}]$
3.	Momentum (P)	$P = mv = \text{Mass} \times \text{Velocity}$	kg - m/s	$[M^1 L^1 T^{-1}]$
4.	Impulse (I)	$I = F \times \Delta t = \text{Force} \times \text{Time}$	Newton-second or kg - m/s	$[M^1 L^1 T^{-1}]$
5.	Force (F)	$F = ma = \text{Mass} \times \text{Acceleration}$	Newton	$[M^1 L^1 T^{-2}]$
6.	Pressure (P)	$P = \frac{F}{A} = \frac{\text{Force}}{\text{Area}}$	Pascal	$[M^1 L^{-1} T^{-2}]$
7.	Kinetic energy (K _E)	$K = \frac{1}{2}mv^2 = \frac{1}{2} \text{Mass} \times \text{Velocity}^2$	Joule	$[M^1 L^2 T^{-2}]$
8.	Power (P)	$P = \frac{W}{t} = \frac{\text{Work}}{\text{Time}}$	Watt or Joule-sec	$[M^1 L^2 T^{-3}]$
9.	Density (d)	$\rho = \frac{m}{V} = \frac{\text{Mass}}{\text{Volume}}$	kg/m ³	$[M^1 L^{-3} T^0]$
10.	Angular displacement (θ)	$\theta = \frac{S}{r} = \frac{\text{Arc}}{\text{Length}}$	Radian (rad)	$[M^0 L^0 T^0]$
11.	Angular velocity (ω)	$\omega = \frac{\Delta \theta}{\Delta t} = \frac{\text{Angular displacement}}{\text{Time}}$	Radian/sec	$[M^0 L^0 T^{-1}]$
12.	Angular acceleration (α)	$\alpha = \frac{\Delta \omega}{\Delta t} = \frac{\text{Angular velocity}}{\text{Time}}$	Radian/sec ²	$[M^0 L^0 T^{-2}]$
13.	Moment of inertia (I)	$I = mx^2 = \text{Mass} \times \text{Distance}^2$	kg-m ²	$[M^1 L^2 T^0]$

S.N.	Quantity	Formula	Unit	Dimension
14.	Torque (τ)	$\tau = F \times r_{\perp} = \text{Force} \times \text{Perpendicular distance}$	Newton-meter	$[M^1L^2T^{-2}]$
15.	Angular momentum (L)	$L = mvr = \text{Mass} \times \text{Velocity} \times \text{Radius}$	Joule-sec	$[M^1L^2T^{-1}]$
16.	Force constant or spring constant (k)	$F = -kx = \text{Force constant} \times \text{Displacement}$	Newton/m	$[M^1L^0T^{-2}]$
17.	Gravitational constant (G)	$F = \frac{Gm_1m_2}{r^2}$ $= \frac{\text{Gravitational constant} \times \text{Mass}^2}{\text{Distance}^2}$	N-m ² /kg ²	$[M^{-1}L^3T^{-2}]$
18.	Gas constant (R)	$PV = nRT$ Pressure = Velocity $\times$ Gas constant $\times$ Temperature	Joule/mol-K	$[M^1L^2T^{-2}\theta^{-1}]$
19.	Planck's constant (h)	$E = hv$ Energy = Planck's constant $\times$ Frequency	Joule-s	$[M^1L^2T^{-1}]$
20.	Surface tension (S)	$S = \frac{F}{L} \Rightarrow \text{Surface Tension} = \frac{\text{Force}}{\text{Length}}$	N/m or Joule/m ²	$[M^1L^0T^{-2}]$
21.	Coefficient of viscosity (η)	$\eta = \frac{F}{6\pi ru} = \frac{\text{Force}}{\text{Radius} \times \text{Velocity}}$	kg/m-s	$[M^1L^{-1}T^{-1}]$
22.	Time period (T)	$T = \frac{1}{n} = \frac{1}{\text{Frequency}}$	Second	$[M^0L^0T^1]$
23.	Frequency (n)	$n = \frac{1}{T} = \frac{1}{\text{Time}}$	Hz	$[M^0L^0T^{-1}]$

Dimensions of Quantities Related to Heat, Thermodynamics and Kinetic Theory of Gases

S.N.	Quantity	Formula	Unit	Dimension
1.	Heat (Q)	Energy	Joule	$[ML^2T^{-2}]$
2.	Specific Heat (c)	$c = \frac{Q}{m \times \Delta\theta} = \frac{\text{Heat}}{\text{Mass} \times \text{Temperature}}$	Joule/kg-K	$[M^0L^2T^{-2}K^{-1}]$
3.	Thermal capacity (K)	$K = \frac{Q}{\Delta t} = \frac{\text{Heat}}{\text{Time}}$	Joule/K	$[M^1L^2T^{-2}K^{-1}]$
4.	Latent heat (L)	$L = \frac{Q}{m} = \frac{\text{Heat}}{\text{Mass}}$	Joule/kg	$[M^0L^2T^{-2}]$
5.	Boltzmann constant (k)	$k = \frac{2E}{3T} = \frac{\text{Energy}}{\text{Temperature}}$	Joule/K	$[M^1L^2T^{-2}K^{-1}]$

S.N.	Quantity	Formula	Unit	Dimension
6.	Coefficient of thermal conductivity (k)	$k = \frac{Qd}{A \times \Delta\theta \times t}$ $= \frac{\text{Heat} \times \text{Distance}}{\text{Area} \times \text{Temp. difference} \times \text{Time}}$	Joule/m-s-K	$[M^1L^1T^{-3}K^{-1}]$
7.	Stefan's constant (σ)	$\sigma = \frac{\Delta E}{A \times \Delta t \times \theta^4}$ $= \frac{\text{Energy}}{\text{Area} \times \text{Time} \times \text{Temperature}^4}$	Watt/m ² -K ⁴	$[M^1L^0T^{-3}K^{-4}]$
8.	Wien's constant (b)	$b = \lambda_{\max} \times T = \text{Wavelength} \times \text{Temperature}$	Meter-K	$[M^0L^1T^0K^1]$
9.	Coefficient of linear expansion (α)	$\alpha = \frac{\Delta L}{L} \frac{1}{T} = \frac{\text{Change in length}}{\text{Length} \times \text{Temperature}}$	Kelvin ⁻¹	$[M^0L^0T^0K^{-1}]$
10.	Mechanical eq. of Heat (J)	$J = \frac{W}{Q} = \frac{\text{Work}}{\text{Heat}}$	Joule/Calorie	$[M^0L^0T^0]$
11.	Vander wall's constant (a)	$a = \frac{RTV^2}{V-b} - PV^2$	Newton-m ⁴	$[M^1L^5T^{-2}]$
12.	Vander wall's constant (b)	Same as Volume (V)	m ³	$[M^0L^3T^0]$
13.	Temperature (T)	$T = \frac{Q}{M\Delta C}$	Kelvin (K)	$[M^0L^0T^0K^1]$

Electricity & Magnetism

S.N.	Quantity	Formula	Unit	Dimension
1.	Electric charge (q)	$q = I \times t = \text{Electric current} \times \text{Time}$	Coulomb	$[M^0L^0T^1A^1]$
2.	Electric current (i)	$I = \frac{q}{t} = \frac{\text{Charge}}{\text{Time}}$	Ampere	$[M^0L^0T^0A^1]$
3.	Capacitance (C)	$C = \frac{q}{V} = \frac{\text{Charge}}{\text{Voltage difference}}$	Coulomb/volt or Farad	$[M^{-1}L^{-2}T^4A^2]$
4.	Electric potential (V)	$V = \frac{q}{C} = \frac{\text{Charge}}{\text{Capacitance}}$	Joule/coulomb	$[M^1L^2T^{-3}A^{-1}]$
5.	Permittivity of free space (ϵ)	$\frac{\text{Charge}^2}{4\pi \times \text{Electric force} \times \text{Distance}^2}$	$\frac{\text{Coulomb}^2}{\text{Newton} - \text{meter}^2}$	$[M^{-1}L^{-3}T^4A^2]$
6.	Dielectric constant (K)	$K = \frac{\epsilon}{\epsilon_0} = \frac{\text{Permittivity in medium}}{\text{Permittivity in free space}}$	Unitless	$[M^0L^0T^0]$

S.N.	Quantity	Formula	Unit	Dimension
7.	Resistance (R)	$R = \frac{V}{I} = \frac{\text{Voltage difference}}{\text{Electric current}}$	Volt/Ampere or Ohm	$[M^1 L^2 T^{-3} A^{-2}]$
8.	Resistivity or Specific resistance (ρ)	$\rho = \frac{RA}{L} = \frac{\text{Resistance} \times \text{Area}}{\text{Length}}$	Ohm-meter	$[M^1 L^3 T^{-3} A^{-2}]$
9.	Coefficient of self-induction (L)	$L = \frac{\mu_0 N^2 A}{I}$	$\frac{\text{Volt} - \text{Second}}{\text{Ampere}}$ or Henry Ohm-second	$[M^1 L^2 T^{-2} A^{-2}]$
10.	Magnetic flux (ϕ)	$\phi = B \times A = \text{Magnetic field} \times \text{Area}$	Volt-second or Weber	$[M^1 L^2 T^{-2} A^{-1}]$
11.	Magnetic induction (B)	$B = \frac{F}{q \times v} = \frac{\text{Magnetic force}}{\text{Charge} \times \text{Velocity}}$	$\frac{\text{Newton}}{\text{Ampere} - \text{meter}}$ or Tesla	$[M^1 L^0 T^{-2} A^{-1}]$
12.	Magnetic Intensity (H)	$H = \frac{B}{\mu} = \frac{\text{Magnetic field}}{\text{Permeability}}$	Ampere/meter	$[M^0 L^{-1} T^0 A^1]$
13.	Magnetic dipole moment (M)	$M = I \times A = \text{Current} \times \text{Area}$	Ampere-meter ²	$[M^0 L^2 T^0 A^1]$
14.	Permeability of free space (μ_0)	$\mu_0 = \frac{B \cdot \ell}{I} = \frac{\text{Magnetic field} \times \text{Length}}{\text{Current}}$	$\frac{\text{Newton}}{\text{Ampere}^2}$	$[M^1 L^1 T^{-2} A^{-2}]$
15.	Surface charge density (σ)	$\sigma = \frac{q}{A} = \frac{\text{Charge}}{\text{Area}}$	Coulomb-meter ²	$[M^0 L^{-2} T^0 A^1]$
16.	Electric dipole moment (p)	$p = q \times d = \text{Charge} \times \text{Distance}$	Coulomb-meter	$[M^0 L^1 T^1 A^1]$
17.	Conductance (G)	$G = \frac{1}{R} = \frac{1}{\text{Resistance}}$	Ohm ⁻¹	$[M^{-1} L^{-2} T^3 A^1]$
18.	Conductivity (σ)	$\sigma = \frac{1}{\rho} = \frac{1}{\text{Resistivity}}$	Ohm ⁻¹ meter ¹	$[M^{-1} L^{-3} T^3 A^2]$
19.	Current density (J)	$J = \frac{I}{A} = \frac{\text{Current}}{\text{Area}}$	Ampere/m ²	$[M^0 L^{-2} T^0 A^1]$
20.	Intensity of electric field (E)	$E = \frac{F}{q} = \frac{\text{Electric force}}{\text{Electric charge}}$	Volt/meter, Neuton/ coulomb	$[M^1 L^1 T^{-3} A^{-1}]$
21.	Rydberg constant (R)	$R_H = \frac{me^4}{8h^3 c \epsilon_0^2}$	m ⁻¹	$[M^0 L^{-1} T^0]$

Quantities Having Same Dimensions

S.N.	Dimension	Quantity
1.	$[M^0L^0T^{-1}]$	Frequency, Angular frequency, Angular velocity and Velocity gradient
2.	$[M^1L^2T^{-2}]$	Work, Internal energy, Potential energy, Kinetic energy. Torque
3.	$[M^1L^{-1}T^{-2}]$	Pressure, Stress, Young's modulus, Bulk modulus, Modulus of rigidity, Energy density
4.	$[M^1L^1T^{-1}]$	Momentum, Impulse
5.	$[M^1L^1T^{-2}]$	Thrust, Force, Weight
6.	$[M^1L^2T^{-1}]$	Angular momentum and Planck's constant
7.	$[M^1L^0T^{-2}]$	Surface tension, Surface energy (energy per unit area), Force constant and Spring constant
8.	$[M^0L^2T^{-2}]$	Latent heat and Gravitational potential
9.	$[M^1L^2T^{-2}\theta^{-1}]$	Thermal capacity, Gas constant and Entropy
10.	$[M^0L^0T^{-1}]$	$L/R, \sqrt{LC}, RC$ where L = Inductance, R = Resistance, C = Capacitance
11.	$[M^0L^1T^0]$	Distance, Displacement, Radius, Wavelength, radius of gyration.
12.	$[M^0L^1T^{-1}]$	Speed, Velocity, Velocity of light.
13.	$[M^0L^1T^{-2}]$	Acceleration, Acceleration due to gravity, Centripetal acceleration.
14.	$[M^0L^0T^{-1}]$	Decay constant, Rate of disintegration.

Units of Some Physical Quantities in Different Systems

Type of Physical Quantity	Physical Quantity	CGS (Originated in France)	MKS (Originated in France)	FPS (Originated in Britain)
Fundamental	Length	cm	m	ft
	Mass	g	kg	lb
	Time	s	s	s
Derived	Force	dyne	newton(N)	poundal
	Work or Energy	erg	joule(J)	ft-poundal
	Power	erg/s	watt(W)	ft-poundal/s

Units and Dimensions of Physical Quantities

Quantity	Common Symbol	SI unit	Dimension
Displacement	s	METRE (m)	L
Mass	m, M	KILOGRAM (kg)	M
Time	t	SECOND (s)	T
Area	A	m^2	L^2
Volume	V	m^3	L^3
Density	ρ	kg/m^3	M/L^3
Velocity	v, u	m/s	L/T
Acceleration	a	m/s^2	L/T^2
Force	F	newton (N)	ML/T^2
Work	W	joule (J) (= N-m)	ML^2/T^2
Energy	E, U, K	joule (J)	ML^2/T^2
Power	P	watt (W) (= J/s)	ML^2/T^3
Momentum	p	$kg\cdot m/s$	ML/T
Gravitational constant	G	$N\cdot m^2/kg^2$	L^3/MT^2
Angle	θ, ϕ	radian	
Angular velocity	ω	radian/s	T^{-1}
Angular acceleration	α	radian/s ²	T^{-2}
Angular momentum	L	$Kg\cdot m^2/s$	ML^2/T
Moment of inertia	I	$Kg\cdot m^2$	ML^2
Torque	τ	N-m	ML^2/T^2
Angular frequency	ω	radian/s	T^{-1}
Frequency	ν	hertz (Hz)	T^{-1}
Period	T	s	T
Young's modulus	Y	N/m^2	M/LT^2
Bulk modulus	B	N/m^2	M/LT^2
Shear modulus	η	N/m^2	ML/T^2
Surface tension	S	N/m	M/T^2
Coefficient of viscosity	η	$N\cdot s/m^2$	M/LT

Pressure	P, p	N/m ² , Pa	M/LT ²
Wavelength	λ	m	L
Intensity of wave	I	W/m ²	M/T ³
Temperature	T	KELVIN (K)	K
Specific heat capacity	c	J/kg-K	L ² /T ² K
Stefan's constant	σ	W/m ² -K ⁴	M/T ³ K ⁴
Heat	Q	J	ML ² /T ²
Thermal conductivity	K	W/m-K	ML/T ³ K

Sach-sach batao, kya tum har physical quantity ka unit aur dimensional formula rat rahe ho na?



Haa humne RATT liye



Nhi RATTNA tha

Poore do saal yahi padhna hai ki kaun-si physical quantity kya hai. So, is par time bilkul waste na karein aur mast rahein. Agar yaad karna hi hai toh velocity, momentum, acc, force, work, power, kinetic energy yaad kar sakte ho.



PRINCIPLE OF HOMOGENEITY

In a physical equation, all terms which have been subtracted or added or equal to must have same dimension. In following equation A, B and C should have same dimension.

$$\begin{cases} A = B + C \\ A = B - C \end{cases} \left\{ \begin{array}{l} A, B, C \text{ are of same DF} \end{array} \right.$$

$$\star P = QR + S \Rightarrow \text{DF of } P, QR, S \text{ will be same}$$

$$\star \text{Change in velocity} = \text{Final velocity} - \text{Initial velocity}$$

$$\Delta \vec{V} = \vec{V}_f - \vec{V}_i$$

Same D.F

$$\star \text{Similarly, in the equation } A = B + C + D \text{ all terms have same dimensions.}$$

$$A = B + C + D$$

D.F. same

Points to be Remember for Dimension Analysis

- ★ Quantity of different dimension cannot be added or subtracted but they can be multiplied or divided.
- ★ Each additive component on the LHS and RHS must have the same dimensional formula this is known as **principle of homogeneity** of equation.
- ★ All trigonometric ratio, exponential and logarithm terms are dimensionless.
- ★ If a physical quantity is multiplied by a number then dimensional formula of that physical quantity remains same.

Q. Sir agar do physical quantity ki dimension Same hai toh kya hum add ya Subst. kar sakte hai ?

Sol. Jaruri nahi for example work and torque have same dimension but work is a scalar and torque is vector quantity so, they can't be added or subtracted

Work, torque $\rightarrow$ D.F. same

Add nahi kar sakte

★ Strain = $\frac{\Delta l}{l} = \frac{\text{Change in Length}}{\text{Original length}}$

★ $\left[\frac{\Delta l}{l}\right] = \left[\frac{\text{Change in Length}}{\text{Original length}}\right] \Rightarrow \frac{L}{L} = 1$

Strain is both unitless and dimensionless

$\Rightarrow [M^0 L^0 T^0]$

Q. What is the unit and dimension of plane angle?

Sol. Angle $\theta = \frac{\text{Arc length}}{\text{Radius}}$

$\theta = \frac{\text{Arc}}{R}$

Dimension $\rightarrow \frac{L}{L} = 1 = [M^0 L^0 T^0]$

Dimensionless

Unit $\rightarrow$ radian, rad

Q. If a physical quantity is dimension-less then it must be unitless. Is true or false?

Sol. False Ex. angle, Solid angle

Application of Dimensional Analysis

- ★ To find the D.F of unknown Physical Quantity.
- ★ We can check the correctness of any formula dimensionally.
- ★ We can derive the relation b/w Physical quantity.
- ★ We can convert from One system of units to other.



Is chapter se majority of the questions upar 4 type of application se puche jaate hain.

TO FIND THE D.F OF UNKNOWN PHYSICAL QUANTITY

Q. If Gravitational force is given by $F = \frac{GM_1 M_2}{r^2}$

then find dimensional formula (D.F) of Gravitational constant $G = ?$

Sol. $[G] = \left[\frac{Fr^2}{m_1 m_2} \right] \Rightarrow \left[\frac{MLT^{-2}L^2}{MM} \right] \equiv [M^{-1}L^3T^{-2}]$

Q. Find dimensional formula of coefficient of viscosity viscous force is given by $F = 6\pi\eta r v$
 $r \rightarrow$ radius, $v \rightarrow$ velocity

Sol. $\eta = \frac{F}{6\pi r v} \Rightarrow [\eta] = \left[\frac{F}{6\pi r v} \right] = \left[\frac{MLT^{-2}}{LLT^{-1}} \right] = [ML^{-1}T^{-1}]$

$6\pi \rightarrow$ Const dimensionless

Q. Dimensional formula of specific heat capacity (S).

Sol. Heat given $\rightarrow$ Rise in temp.

$\Delta Q = MS\Delta T$

Specific heat capacity

$S = \frac{\Delta Q}{M\Delta T} \Rightarrow [S] = \left[\frac{\Delta Q}{M\Delta T} \right] = \left[\frac{ML^2T^{-2}}{MK} \right] \equiv [M^0 L^2 T^{-2} K^{-1}]$

Q. Find dimensional formula (D.F) of Boltzman Constant (k). If formula of internal energy is given by

$U = \frac{3}{2} KT$
 Kinetic Energy (Internal Energy) $\rightarrow$ temp

Sol. From above formula Boltzman Constant (k)

$[K] = \left[\frac{2U}{3T} \right] \rightarrow$ temp

$[K] \Rightarrow \left[\frac{ML^2T^{-2}}{K} \right] \equiv [ML^2T^{-2}K^{-1}]$
 Boltzman Constant $\rightarrow$ D.F. of temp

Q. Find dimensional formula of charge if formula of charge is equal to current $\times$ time

Sol. $q = i\Delta t$

D.F of charge = $[AT] \equiv [M^0 L^0 TA]$

Q. Find the dimensional formula of electric potential or voltage

$$\text{Electric potential} = \frac{\text{Electric Potential Energy}}{\text{Charge}}$$

Sol. Dimension of electric potential is $= \frac{ML^2T^{-2}}{AT}$
 $= [ML^2T^{-3}A^{-1}]$

Q. Find the dimensional formula of resistance

We know $V = iR$
 Pot. Difference Resistance

Sol. The formula of resistance is

$$= [R] = \left[\frac{V}{i} \right] = \frac{[ML^2T^{-3}A^{-1}]}{[A]}$$

$$= \frac{[ML^2T^{-3}A^{-1}]}{[A]} = [ML^2T^{-3}A^{-2}]$$

Q. If $R = \rho \frac{\ell}{(\text{Area})}$ where R is resistance, ℓ is length and ρ is resistivity. Find DF of resistivity.

Sol. $\rho = \frac{R \times (\text{Area})}{\ell}$
 $\rho = \frac{ML^2T^{-3}A^{-2} \cdot L^2}{L} \Rightarrow [ML^3T^{-3}A^{-2}]$

Q. Kinetic energy of a particle moving along elliptical trajectory is given by $K = \alpha s^2$ where s is the distance travelled by the particle. Determine dimensions of α .

Sol. $K = \alpha s^2 \Rightarrow \alpha = \frac{K}{s^2}$
 $[\alpha] = \left[\frac{(ML^2T^{-2})}{(L^2)} \right]$
 $[\alpha] = [M^1 L^0 T^{-2}]$

We can Check the Correctness of any Formula Dimensionally

Q. Check dimensionally if this formula $F = \frac{Mv^2}{r^3}$ is correct? (Where v is speed r is radius, F is force, and M is mass)

Sol. DF of LHS = Force $\Rightarrow [ML^1T^{-2}]$
 DF of RHS $\Rightarrow \frac{mv^2}{r^3} \Rightarrow \frac{M(LT^{-1})^2}{L^3} = [ML^{-1}T^{-2}]$
 Since D.F. of LHS $\neq$ D.F. of RHS so given formula is wrong

Q. Is this formula $v = \sqrt{\frac{GM}{r}}$ correct or wrong where v is orbital velocity, M is mass of earth, R is radius of orbit and G is gravitational constant

Sol. In this formula DF of LHS $\Rightarrow [v] \Rightarrow [LT^{-1}]$

$$\text{RHS} \Rightarrow \sqrt{\frac{GM}{r}} = \sqrt{\frac{M^{-1}L^3T^{-2}M}{L}} = \sqrt{L^2T^{-2}} = LT^{-1}$$

Since D.F. of LHS = D.F. of RHS so given formula is dimensionally correct.

Q. Check whether the given relation $\frac{Fv^2}{t} = KE$ (Kinetic Energy) is dimensionally correct? Here F = force, v = velocity and t = time?

Sol. $[\text{LHS}] = \left[\frac{Fv^2}{t} \right] = \frac{[MLT^{-2}] \times [L^2T^{-2}]}{[T]} = [ML^3T^{-5}]$

$$[\text{RHS}] = [KE] = \left[\frac{1}{2}mv^2 \right] = [M \times L^2T^{-2}] = [ML^2T^{-2}]$$

$\therefore [\text{LHS}] \neq [\text{RHS}]$, so the given relation is incorrect dimensionally.

- ★ If a formula is dimensionally incorrect $\Rightarrow$ it must be incorrect overall.
- ★ If a formula is dimensionally correct $\Rightarrow$ Then formula may be correct may be incorrect but अभी हम only dimensionally correctness पर बात करेंगे।
- ★ If a formula is correct then it must be both dimensionally correct and numerically correct.

Q. The distance covered by a particle in time t is given by $x = a + bt + ct^2 + dt^3$; find the dimensions of a , b , c and d .

Sol. The equation contains five terms. All of them should have the same dimensions. Since $[x] =$ length, each of the remaining four must have the dimension of length.

Thus, $[a] = \text{length} = L$

$$[bt] = L, \quad \text{or } [b] = [LT^{-1}]$$

$$[ct^2] = L, \quad \text{or } [c] = [LT^{-2}]$$

$$\text{and } [dt^3] = L \quad \text{or } [d] = [LT^{-3}]$$

Q. If x is displacement t is time in the equation $x = \frac{A}{t} + Bx^2 + Cx$. Find the dimensional formula of A , B and C .

Sol. $[x] = \left[\frac{A}{t} \right] + [Bx^2] + [Cx]$

Dimension formula of $\frac{A}{t}$ = Dimensional formula of x .

$$\left[\frac{A}{t} \right] = [L] \Rightarrow A = LT$$

$$[BL^2] = [L] \Rightarrow [L] \Rightarrow [B] = [L^{-1}]$$

$$[CL] = [L] \text{ So, } C = \text{dimensionless}$$

Q. If x is displacement t is time in the equation $x = Ax^2 + \frac{B}{t^2} + \frac{C}{x}$. Find the dimensional formula of A , B and C .

If D.F. of $\frac{AB}{C}$ is $M^\alpha L^\beta T^\gamma$ find $\alpha + \beta + \gamma$

Sol. $x = Ax^2 + \frac{B}{t^2} + \frac{C}{x}$

From principle of homogeneity

$$[x] = [Ax^2] = \left[\frac{B}{t^2} \right] = \left[\frac{C}{x} \right]$$

$$(i) [Ax^2] = [x]$$

$$\therefore AL^2 = L, \quad [A] = [L^{-1}]$$

$$(ii) \left[\frac{B}{t^2} \right] = [x]$$

$$\therefore \frac{B}{T^2} = L, \quad [B] = [LT^2]$$

$$(iii) \left[\frac{C}{x} \right] = [x]$$

$$\frac{C}{L} = L \quad [C] = [L^2]$$

$$\frac{AB}{C} = \frac{L^{-1}LT^2}{L^2} = M^0L^{-2}T^2$$

$$\text{So, } \alpha + \beta + \gamma = 0 + (-2) + 2 = 0$$

Q. If x is displacement t is time and v is velocity in the equation $v = At + \frac{B}{x^2} + Ct^2$. Find the dimensional formula of ABC .

Sol. $[v] = [At] + \left[\frac{B}{x^2} \right] + [Ct^2]$

$$(i) [v] = [At]$$

$$LT^{-1} = AT$$

$$A = LT^{-2}$$

$$(ii) \left[\frac{B}{x^2} \right] = [v]$$

$$\frac{B}{L^2} = LT^{-1}$$

$$B = L^3T^{-1}$$

$$(iii) [Ct^2] = [v]$$

$$CT^2 = LT^{-1}$$

$$C = LT^{-3}$$

$$\text{So, dimensional formula of } ABC \equiv LT^{-2}L^3T^{-1}LT^{-3} \equiv M^0L^5T^{-6}$$

Q. If x is displacement, v is velocity and t is time in the equation $v = \frac{Ax}{B+t}$. Find the dimensional formula of A and B .

Sol. $[v] = \left[\frac{Ax}{B+t} \right]$

$$\therefore B = T$$

Dimensionally

$$\Rightarrow [v] = \left[\frac{Ax}{\text{time}} \right] \Rightarrow LT^{-1} = \frac{AL}{T}$$

A = any number Dimensionless

Q. If x is displacement, v is velocity in the equation $v = \frac{A}{B-x^2}$. Find the dimensional formula of A and B .

Sol. From homogeneity principle D.F. of B = D.F. of x^2

$$\therefore [B] = [L^2]$$

$$[v] = \left[\frac{A}{L^2} \right] \Rightarrow LT^{-1} = \frac{A}{L^2}$$

$$[A] = [L^3T^{-1}]$$

SKC: Kaddu mein kaddu hi jud sakta hai aur yahan B mein t jud raha hai. So, dimension of B and t are equal.

SKC

Ye dekho, kisme kya jud raha hai ya kisse kya ghat raha hai? Ques (questions) wahi se tutega.



Q. Potential energy (U) is given by $U = \frac{A\sqrt{x}}{x^2 + B}$ x is

distance. Find the dimensional formula of A and B . Also find $[AB]$.

Sol. From principle of homogeneity dimension of L.H.S = dimension of R.H.S

$$[U] = \left[\frac{A\sqrt{x}}{x^2 + B} \right]$$

$$\Rightarrow ML^2T^{-2} = \frac{AL^{\frac{1}{2}}}{L^2} = \frac{A}{L^{\frac{3}{2}}}$$

$$\Rightarrow [A] = [M^1L^{\frac{7}{2}}T^{-2}]$$

Since B is added to x^2

$$[B] = [L^2] \Rightarrow B = L^2$$

D.F. of $\underline{A.B}$ is

$$[AB] \Rightarrow ML^{\frac{7}{2}}T^{-2}L^2 = ML^{\frac{11}{2}}T^{-2}$$

Q. If $\frac{v + A\sqrt{s}}{Bt^2} = F$; find the dimension of A and B where F = force, v = velocity, s = displacement and t = time

Sol. Using principle of homogeneity; $[A\sqrt{s}] = [v]$

$$[AL^{\frac{1}{2}}] = [LT^{-1}]$$

$$\therefore [A] = [L^{\frac{1}{2}}T^{-1}]$$

$$\text{Also, we can write, } B = \frac{v + A\sqrt{s}}{Ft^2}$$

$$[B] = \frac{[LT^{-1}]}{[(MLT^{-2})(T^2)]} \Rightarrow [B] = [M^{-1}L^0T^{-1}]$$

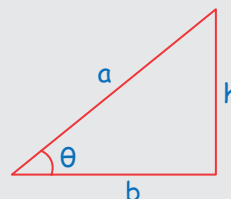
Important Points

★ Angle is dimensionless $\Rightarrow [\theta]$ is dimensionless because angle $\theta = \text{arc}/\text{radius}$

★ Sin (kaddu) here sin ke andar angle hoga, so (kaddu) is Dimensionless.

★ Sin (kaddu) is also Dimensionless because

$$\sin \theta = \frac{h}{a} \equiv \text{dimensionless}$$



★ All the trigonometry ratio $\sin \theta$, $\cos \theta$, $\tan \theta$ etc. are dimensionless.

★ Log (kaddu) and In (kaddu)

Dimensionless

★ Inverse function is also dimensionless

★ $\ln x = \sin^{-1}(y)$ here x and y both are dimensionless.

★ $y = \log x \longrightarrow$ here x and y are dimensionless

★ $e^x = y \longrightarrow$ here x and y are dimensionless

★ $a^x = y \longrightarrow$ here x and y are dimensionless

★ Power of exponential function is dimensionless

Dekh bhai, samajh raha hai na? Mudde ki baat ye hai ki sin ke andar, cos ke andar, tan ke andar, sin inverse ke andar, log ke andar, e ki power mein jo bhi hai, woh dimensionless hoga.



Ex. In $y = \sin x \longrightarrow$ Displacement mat samajhna
 $\downarrow$
 Dimensionless

Here y is number between $[-1, 1]$ because range of sin is between $[-1, 1]$

Q. In equation $y = \sin (At^2)$. Find the dimension of A .

Sol. At^2 is dimensionless

$$\Rightarrow At^2 = M^0L^0T^0 = 1$$

$$\Rightarrow A = T^{-2}$$

Q. In equation $y = \sin(At^3 + Bx^2)$. Find the dimension of A and B.

Sol. $[At^3] + [Bx^2]$ this will be dimensionless

$$\Rightarrow AT^3 = 1 \text{ and } BL^2 = 1$$

$$A = T^{-3} \quad B = \frac{1}{L^2}$$

Q. In equation $y = \sin\left(\frac{A}{t^2} + Bx^2\right)$. Find the dimension of A or B.

Sol. Argument of sine are dimensionless

$$\left[\frac{A}{t^2}\right] \Rightarrow \frac{A}{T^2} = 1 \Rightarrow A = T^2$$

$$\text{and } [Bx^2] = 1 \Rightarrow BL^2 = 1 \Rightarrow B = L^{-2}$$

Ab main kya bolu, ab tak tu khud samajh gaya hoga ki sin ke andar jitne individual terms hain unko 1 ke barabar karna hai. Same thing for cos, log, e. Abe, samajh toh gaya na?



Q. In equation $y = \log\left(\frac{A}{t^2} + Bx^2\right)$ Find the dimension of A and B.

Sol. Argument of log is dimensionless so

$$\left(\frac{A}{t^2} + Bx^2\right) \text{ is dimensionless}$$

$$\frac{A}{T^2} = 1 \text{ and } [Bx^2] = 1 \Rightarrow BL^2 = [1]$$

Q. In equation $y = e^{\frac{A}{t^2} + Bx^2}$. Find the dimension of A and B.

Sol. Same as above ques.

Q. In equation $y = A \sin(Bt + C)$, given A is amplitude in meter. Find the dimension of B and C.

Sol. Argument of trigonometric function are dimensionless

$$\Rightarrow C = 1$$

$$\text{and } BT = 1$$

$$B = T^{-1}$$

$\sin(Bt + C)$ is also dimensionless with Value $[-1, 1]$

$$\text{D.F. of } y = L^1 \\ y = m^0 L^1 T^0$$

Q. In equation $y = A \sin\left(\frac{B}{t^2} + cv\right)$, given A is amplitude in meter v is velocity. Find the dimension of B and C.

$$\text{Sol. } \left[\frac{B}{t^2} + cv\right] = 1$$

$$\Rightarrow \frac{B}{T^2} = 1 \Rightarrow B = T^2$$

$$\text{and } c \cdot v = 1$$

$$c \cdot LT^{-1} = 1$$

$$[c] = [L^{-1} T]$$

$$[y] = [m^0 L^1 T^0]$$

Q. In equation $q = Q_0 e^{-t/\tau}$ find D.F. of τ .

Sol. $\frac{t}{\tau}$ will be dimensionless because power of exponential function is dimensionless.

$$\frac{t}{\tau} = 1 \Rightarrow [\tau] = [T]$$

Q. In equation $y = A e^{Bx^2 + Cx + Dt^3}$ Find the dimensional formula of B, C and D.

Sol. Dimension of y will be same as that of A

$$y = m^0 L^1 T^0$$

power of e will be dimensionless so

$$[Bx^2] = [Cx] = [Dt^3] = [1]$$

$$[B] = [L^{-2}]$$

$$[C] = [L^{-1}]$$

$$[D] = [T^{-3}]$$

Q. In the equation pressure (P) is given by formula

$$P = \frac{\alpha}{\beta} e^{\frac{\alpha Z}{k\theta}} \text{ where } Z \text{ is distance } P \text{ is pressure } k \text{ is}$$

Boltzmann constant θ is temperature. Find the dimension formula of α and β .

Sol. Internal energy U is given by $U = 3/2 kT$ where T is temperature $\Rightarrow [kT] = [U]$

$$[k\theta] = [ML^2 T^{-2}]$$

$$[\alpha Z] [\alpha L] = [ML^2 T^{-2}]$$

$$\alpha = MLT^{-2}$$

D.F. of $P = \text{D.F. of } \frac{\alpha}{\beta}$

$$\frac{MLT^{-2}}{L^2} = \frac{MLT^{-2}}{\beta}$$

$$\beta = L^2$$

Q. The position of a particle at time t , is given by the equation, $x(t) = \frac{v_0}{\alpha} (1 - e^{-\alpha t})$, where v_0 is a constant and $\alpha > 0$. The dimensions of v_0 and α are respectively.

Sol. $[a][t] = [M^0 L^0 T^0]$ and $[v_0] = [x][a]$
 $[a] = [M^0 L^0 T^{-1}] = [M^0 L^1 T^{-1}]$

WE CAN DERIVE THE RELATION BETWEEN PHYSICAL QUANTITY

Bhai, is prakar ke question ki physics do line ki, but maths 10 line ki bhi ho sakti hai. All questions are of same pattern. Let's try to understand.



Q. Time period of simple pendulum depends on mass (m) of block, length (l) of simple pendulum, acc. due to gravity (g) Obtain the formula of time period (T) dimensionally.

Sol. Suppose $T \propto m^x$
 $T \propto l^y$ and k is dimensionless constant

$$\frac{T \propto g^z}{T \propto K m^x l^y g^z}$$

Using principle homogeneity dimension of L.H.S must be equal to dimension R.H.S

$$T = K m^x l^y g^z$$

$$[LHS] \rightarrow [M^0 L^0 T^1]$$

$$[RHS] = [M^x L^y g^z]$$

$$[L.H.S] = [R.H.S]$$

$$M^0 L^0 T^1 = M^x L^y (LT^{-2})^z$$

$$M^0 L^0 T^1 = M^x L^{y+z} T^{-2z}$$

Comparing powers of M, L, T in L.H.S and R.H.S

$$x = 0, \quad y + z = 0$$

$$-2z = 1$$

$$z = -\frac{1}{2}$$

$$y = +\frac{1}{2}$$

Putting values of x, y, z in the formula of time period T we get

$$T = k \sqrt{\frac{l}{g}}$$

$$T = k m^0 l^{\frac{1}{2}} g^{-\frac{1}{2}}$$

$$\text{Actual} \rightarrow T = 2\pi \sqrt{\frac{l}{g}}$$

Ques ki language ke matlab par nahi jaana hai, na hi physics ko samajhne ki koshish karni hai. Bas ye dekhna hai ki is tarah ke sawal mathematically kaise solve ho rahe hain aur eqn (equation) solve karte waqt calculation mistake nahi honi chahiye. Isi baat pe next ques khud se solve karo.



Q. A gas bubble oscillate with time period T proportional to P (pressure), d (density), and E (energy) find relation between them using dimension analysis.

$$[\text{Pressure}(P)] = \left[\frac{MLT^{-2}}{L^2} \right]$$

Sol.

Sir ye oscillation kya hota hai



Ye sab hum mechanics mein padhenge, abhi iska load mat lo.

Suppose time period varies with p, d and E by the following relation

$$T = K p^x d^y E^z$$

Using principle of homogeneity

$$[T] = [K p^x d^y E^z] \quad K \text{ is dimensionless}$$

$$M^0 L^0 T^1 = (ML^{-1} T^{-2})^x \cdot (ML^{-3})^y (ML^2 T^{-2})^z$$

$$M^0 L^0 T^1 = M^{x+y+z} L^{-x-3y+2z} T^{-2x-2z}$$

$$-2x - 2z = 1$$

$$x + y + z = 0$$

$$-x - 3y + 2z = 0$$

Solve yourself

Q. A satellite is orbiting around a planet, its orbiting velocity depends on R (radius of orbit), m (mass of planet) and G (gravitational constant). Using dimensional analysis find an expression relating orbital velocity of satellite

Sol. Suppose orbital velocity (v) depends on R , M and G by the following relation

$$v = KR^x m^y G^z \quad K \text{ is dimensionless constant}$$

Using principle homogeneity

$$[v] = [KR^x m^y G^z]$$

$$LT^{-1} = L^x M^y (M^{-1} L^3 T^{-2})^z$$

$$M^0 L^1 T^{-1} = M^{y-z} L^{x+3z} T^{-2z}$$

Comparing powers of M , L , T on both sides of the equation we get

$$-2z = -1 \quad \dots (i) \quad \Rightarrow z = \frac{1}{2}$$

$$y - z = 0, \dots (ii) \quad y = z$$

$$\Rightarrow y = \frac{1}{2}, \quad z = \frac{1}{2}, \quad x = -\frac{1}{2}$$

$$x + 3z = 1 \quad (iii) \quad y = \frac{1}{2}$$

$$x + \frac{3}{2} = 1 \quad z = \frac{1}{2}$$

$$x = -\frac{1}{2}$$

putting values of x , y , z in $v = KR^x m^y G^z$ we get

$$v = kR^{-\frac{1}{2}} m^{\frac{1}{2}} G^{\frac{1}{2}}$$

$$v = k \sqrt{\frac{Gm}{R}}$$

Sir, kya in sawalon mein proportionality constant K hamesha dimensionless hoga? Haan bhai, aise sawalon mein K ko dimensionless hi manna hai. Agar K ki dimension hui, toh woh ques (question) is method se solve nahi hoga aur ye ek dimensional analysis ki limitation bhi hai.



Q. Energy of a particle oscillating depends on mass (m) frequency ' f ', and amplitude ' A ' such that

$$E = K m^x f^y A^z \quad \text{find value of } x + y + z$$

dimensionless

$$ML^2 T^{-2} = M^x (T^{-1})^y L^z$$

$$ML^2 T^{-2} = M^x L^z T^{-y}$$

$$x = 1$$

$$z = 2$$

$$y = 2$$

$$1 + 2 + 2 = 5 \text{ Ans.}$$

Q. When a solid sphere moves through a liquid, the liquid opposes the motion with a force F . The magnitude of F depends on the coefficient of viscosity η of the liquid, the radius r of the sphere and the speed v of the sphere. Assuming that F is proportional to different powers of these quantities, guess a formula for F using the method of dimensions.

Sol. Suppose the formula is $F = k \eta^a r^b v^c$

$$\text{Then, } [MLT^{-2}] = [ML^{-1} T^{-1}]^a [L]^b \left[\frac{L}{T} \right]^c$$

$$= [M^a L^{-a+b+c} T^{-a-c}]$$

Equating the exponents of M , L and T from both sides,

$$a = 1$$

$$-a + b + c = 1$$

$$-a - c = -2$$

Solving these, $a = 1$, $b = 1$ and $c = 1$

Thus, the formula for F is $F = k \eta r v$.

UNIT CONVERSION FROM DIMENSION ANALYSIS

Q. Convert 1 Newton into dyne

dyne $\rightarrow$ It is unit of force in CGS system

Sol. The dimensional equation of force is $[F] = [M^1 L^1 T^{-2}]$

Therefore if n_1 , u_1 and n_2 , u_2 corresponds to SI and CGS numerical value and unit respectively, then

$$n_2 = n_1 \left[\frac{M_1}{M_2} \right]^1 \left[\frac{L_1}{L_2} \right]^1 \left[\frac{T_1}{T_2} \right]^{-2}$$

$$= 1 \left[\frac{\text{kg}}{\text{g}} \right] \left[\frac{\text{m}}{\text{cm}} \right] \left[\frac{\text{s}}{\text{s}} \right]^{-2} = 1 \times (1000)(100)(1) = 10^5$$

Method-2:

$$1 \text{ N} = 1 \text{ kgm/sec}^2 \xrightarrow{\text{CGS}} 1 \cdot \frac{1000 \text{ gm} \cdot 100 \text{ cm}}{\text{sec}^2}$$

$$= 10^5 \frac{\text{gm} \cdot \text{cm}}{\text{sec}^2} = 10^5 \text{ (dyne)}$$

Q. Convert 1 Joule into erg

erg → Unit of energy in CGS system

Sol. $1 \text{ Joule} = \frac{1 \text{ kg m}^2}{\text{sec}^2} \Rightarrow \frac{1 \times 1000 \text{ gm} \times (100 \text{ cm})^2}{\text{sec}^2}$

$$= 10^7 \frac{\text{gm cm}^2}{\text{sec}^2}$$

$$= 10^7 \text{ erg}$$

Q. The dimensional formula for viscosity of fluids is $[\eta] = [M^1 L^{-1} T^{-1}]$. Find how many poise (CGS unit of viscosity) is equal to 1 poiseuille (SI unit of viscosity).

Sol. $[\eta] = [M^1 L^{-1} T^{-1}]$

$$1 \text{ CGS units} = \text{g cm}^{-1} \text{ s}^{-1}$$

$$1 \text{ SI units} = \text{kg m}^{-1} \text{ s}^{-1}$$

$$= (1000 \text{ g})(100 \text{ cm})^{-1} \text{ s}^{-1} = 10 \text{ g cm}^{-1} \text{ s}^{-1}$$

Thus, 1 Poiseuille = 10 poise

Q. A calorie is a unit of heat or energy and it equals about 4.2 J, where $1 \text{ J} = 1 \text{ kg m}^2/\text{s}^2$. Suppose we employ a system of units in which the unit of mass equals $\alpha \text{ kg}$, the unit of length equals $\beta \text{ metre}$, the unit of time is $\gamma \text{ second}$. Show that a calorie has a magnitude $4.2 \alpha^{-1} \beta^{-2} \gamma^2$ in terms of the new units.

Sol. $1 \text{ cal} = 4.2 \text{ kg m}^2 \text{ s}^{-2}$

SI New system

$$n_1 = 4.2 \quad n_2 = ?$$

$$M_1 = 1 \text{ kg} \quad M_2 = \alpha \text{ kg}$$

$$L_1 = 1 \text{ m} \quad L_2 = \beta \text{ metre}$$

$$T_1 = 1 \text{ s} \quad T_2 = \gamma \text{ second}$$

Dimensional formula of energy is $[ML^2T^{-2}]$

Comparing with $[M^a L^b T^c]$,

We find that $a = 1, b = 2, c = -2$

$$\text{Now, } n_2 = n_1 \left[\frac{M_1}{M_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$$

$$= 4.2 \left[\frac{1 \text{ kg}}{\alpha \text{ kg}} \right]^1 \left[\frac{1 \text{ m}}{\beta \text{ m}} \right]^2 \left[\frac{1 \text{ s}}{\gamma \text{ s}} \right]^{-2} = 4.2 \alpha^{-1} \beta^{-2} \gamma^2$$

Ab main aapke liye neeche bahut saare attach kar raha hoon, inhe sincerely solve karein khud se, bcz (because) Mains/NEET/Advance sabhi exams mein aise hi question aane ki probability sabse highest hai. Ye lo pen aur start ho jao. Shuru!



Q. In a given system of Units. 1 Unit of mass = 2kg
1 Unit of length = 5m 1 Unit of time = 5sec
Then in this system, 1N represent

Sol. Unit of force in hypothetical system

$$= \frac{2 \text{ kg} \cdot 5 \text{ m}}{(5 \text{ sec})^2} = \frac{10}{25} \frac{\text{kgm}}{\text{sec}^2}$$

$$1 \text{ N}' = \frac{2}{5} \text{ N} \quad \text{हमारा वाला}$$

$$1 \text{ N} = \frac{5}{2} \text{ N}'$$

Q. Unit of force in given system is represented by N' . In this system 1 Unit of mass = 20kg 1 Unit of length = 4m 1 Unit of time = 2sec. Answer following parts of questions.

Sol. (i) 1 Unit of force in this system

$$= \frac{20 \text{ kg} \cdot 4 \text{ m}}{(2 \text{ sec})^2} = \frac{20 \text{ kg m}}{\text{sec}^2} = 20 \text{ N}$$

$$(ii) \text{ Numerical value of } 1 \text{ N in this system} = \frac{1}{20} \text{ N}'$$

(iii) Numerical value of 50N in this system

$$= \frac{50}{20} \text{ N}'$$

**Few Important Question You Must Solve Because
ye exam mein repeatedly puche gye hai**

Q. If P is the pressure of a gas and ρ is its density, then find the dimension of velocity in terms of P and ρ .

Sol. Method - I

$$[P] = [ML^{-1}T^{-2}] \quad \dots(i)$$

$$[\rho] = [ML^{-3}] \quad \dots(ii)$$

Dividing eq. (i) by (ii)

$$[P\rho^{-1}] = [L^2T^{-2}]$$

$$\Rightarrow [LT^{-1}] = [\rho^{1/2}P^{-1/2}]$$

$$\Rightarrow [v] = [P^{1/2}\rho^{-1/2}]$$

Method - II

$$v \propto P^a \rho^b$$

$$v = kP^a \rho^b$$

$$[LT^{-1}] = [ML^{-1}T^{-2}]^a [ML^{-3}]^b$$

$$\Rightarrow a = \frac{1}{2}, b = -\frac{1}{2} \quad (\text{Equating dimensions})$$

$$\Rightarrow [v] = [P^{1/2}\rho^{-1/2}]$$

Q. Find relationship between speed (v) of sound in a medium, the elastic constant (E) and the density of the medium (ρ).

Sol. Let the speed depends upon elastic constant and density according to the relation

$$v \propto E^a \rho^b$$

$$\text{or } v = KE^a \rho^b \quad \dots(i)$$

Where K is a dimensionless constant of proportionality.

Considering dimensions of the quantities

$$[v] = [M^0 L T^{-1}]$$

$$[E] = \frac{[\text{stress}]}{[\text{strain}]} = \frac{[\text{force}]/[\text{area}]}{[\Delta\ell]/[\ell]} = \frac{[M^1 L^1 T^{-2}]/[L^2]}{[L^1]/[L^1]} = [M^1 L^{-1} T^{-2}]$$

$$\therefore [E^a] = [M^a L^{-a} T^{-2a}]$$

$$[\rho] = [\text{mass}]/[\text{volume}] = [M]/[L^3] = [M^1 L^{-3} T^0]$$

$$\therefore [\rho^b] = [M^b L^{-3b} T^0]$$

Equating the dimensions of the LHS and RHS quantities of equation (i), we get

$$[M^0 L^1 T^{-1}] = [M^a L^{-a} T^{-2a}] \times [M^b L^{-3b} T^0]$$

$$\text{or } [M^0 L^1 T^{-1}] = [M^{a+b} L^{-a-3b} T^{-2a}]$$

Comparing the individual dimensions of M , L and T

$$a + b = 0, \quad \dots(ii)$$

$$-a - 3b = 1, \text{ and} \quad \dots(iii)$$

$$-2a = -1 \quad \dots(iv)$$

Solving we get

$$a = \frac{1}{2}, b = -\frac{1}{2}$$

Therefore the required relation is

$$v = K \sqrt{\frac{E}{\rho}}$$

Q. Pressure (P) acting due to a fluid kept in a container depends on, weight of liquid (w), Area of cross-section of container (A) and density of fluid (ρ). Establish a formula of pressure (P).

Sol. According to the question

$$P \propto w^x A^y \rho^z$$

$$P = K[w^x A^y \rho^z]$$

Now writing the dimension of each quantity on either side.

$$[ML^{-1}T^{-2}] = K [MLT^{-2}]^x [L^2]^y [ML^{-3}]^z$$

$$[ML^{-1}T^{-2}] = K [M]^{x+z} [L]^{x+2y-3z} [T]^{-2x}$$

Now comparing the powers

$$\text{For } M, 1 = x + z \quad \dots(i)$$

$$L, -1 = x + 2y - 3z \quad \dots(ii)$$

$$T, -2 = -2x \quad \dots(iii)$$

$$\Rightarrow \text{Solving we get, } x = 1, y = -1, z = 0$$

$$\therefore P = KwA^{-1}\rho^0$$

Q. If ϵ_0 is the permittivity of free space and E is the electric field, then $\epsilon_0 E^2$ has the dimensions:

Ans. $ML^{-1}T^{-2}$

Hint: Energy density = $\frac{\text{Energy}}{\text{Volume}} = \frac{1}{2} \epsilon_0 E^2$

Q. Match List-I with List-II.

List-I		List-II	
A.	Coefficient of viscosity	I.	$[M L^2 T^{-2}]$
B.	Surface Tension	II.	$[M L^2 T^{-1}]$
C.	Angular momentum	III.	$[M L^{-1} T^{-1}]$
D.	Rotational kinetic energy	IV.	$[M L^0 T^{-2}]$

Sol. As we know,

$$F = \eta A \frac{dv}{dy}, L = mvr, S.T = \frac{F}{\ell}, K.E = \frac{1}{2} I \omega^2$$

Hence,

$$F = \eta A \frac{dv}{dy},$$

$$[MLT^{-2}] = \eta [L^2] [T^{-1}]$$

$$\eta = [ML^{-1}T^{-1}]$$

$$S.T = \frac{F}{\ell} = \frac{[MLT^{-2}]}{[L]} = [ML^0T^{-2}]$$

$$L = mvr = [ML^2T^{-1}]$$

$$K.E = \frac{1}{2} I \omega^2 = [ML^2T^{-2}]$$

Q. The equation of state of a real gas is given by

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT, \text{ where } P, V \text{ and } T \text{ are pressure, volume and temperature respectively and } R \text{ is the universal gas constant. The}$$

dimensions of $\frac{a}{b^2}$ is similar to that of:

Sol. $[P] = \left[\frac{a}{V^2}\right] \Rightarrow [a] = [PV^2]$

And dimension of V = dimension of b

$$\therefore \frac{[a]}{[b^2]} = \frac{[PV^2]}{[V^2]} = [P]$$

Q. Young's modulus of elasticity Y is expressed in terms of three derived quantities, namely, the gravitational constant G , Planck's constant h and the speed of light c , as $Y = c^\alpha h^\beta G^\gamma$. Which of the following is the correct option?

Sol. $Y = c^\alpha h^\beta G^\gamma$

$$[ML^{-1}T^{-2}] = [LT^{-1}]^\alpha [ML^2T^{-1}]^\beta [M^{-1}L^3T^{-2}]^\gamma$$

$$[ML^{-1}T^{-2}] = [M^{\beta-\gamma}][L^{\alpha+2\beta+3\gamma}][T^{-\alpha-\beta-2\gamma}]$$

$$1 = \beta - \gamma \quad \dots(i)$$

$$-1 = \alpha + 2\beta + 3\gamma \quad \dots(ii)$$

$$-2 = -\alpha - \beta - 2\gamma \quad \dots(iii)$$

After solving above equations

$$\alpha = 7, \beta = -1, \gamma = -2$$

Q. In a particular system of units, a physical quantity can be expressed in terms of the electric charge e , electron mass m_e , Planck's constant h , and coulomb's constant

$$k = \frac{1}{4\pi\epsilon_0}, \text{ where } \epsilon_0 \text{ is the permittivity}$$

of vacuum. In terms of these physical constants, the dimension of the magnetic field is $[B] = [e]^\alpha [m_e]^\beta [h]^\gamma [k]^\delta$. The value of $\alpha + \beta + \gamma + \delta$ is _____.

Sol. $MT^{-2}I^{-1} = [IT]^\alpha [M]^\beta [ML^2T^{-1}]^\gamma [ML^3T^{-4}I^{-2}]^\delta$

$$\therefore \beta + \gamma + \delta = 1 \quad \dots(i)$$

$$2\gamma + 3\delta = 0 \quad \dots(ii)$$

$$\alpha - \gamma - 4\delta = -2 \quad \dots(iii)$$

$$\alpha - 2\delta = -1 \quad \dots(iv)$$

On solving these equations, we get

$$\alpha + \beta + \gamma + \delta = 4$$

Q. If time (t), velocity (v), and angular momentum (ℓ) are taken as the fundamental units. Then the dimension of mass (m) in terms of t, v , and ℓ is:

Sol. Let us suppose,

$$\text{Mass } \alpha [t^a v^b \ell^c] \Rightarrow [M^1 L^0 T^0] = T^a \times [LT^{-1}]^b \times [ML^2 T^{-1}]^c$$

On equating both sides, we get

$$\Rightarrow a - b - c = 0; c = 1, \text{ and } b + 2c = 0 \Rightarrow b = -2c = -2$$

$$\Rightarrow a = b + c = 1 - 2 = -1$$

$$\text{Hence, dimension of mass} = [t^a v^b \ell^c] = [t^{-1} v^{-2} \ell^1]$$

Q. In a hypothetical system if 1 Unit of mass = 10 kg 1 Unit of length = 5 m 1 Unit of sec = 2 sec.

Find

(1) 1 unit of force in this hypothetical system in terms of newton.

(2) 1 Unit of Energy in this hypothetical system 1 Unit of force in this system

(3) Find numerical value of 500 Joule in terms of new system

Sol. SKC Method

(i) Value of 1 unit of force in hypothetical

$$\text{system} = 1 N' = \frac{10 \text{ kg} \times 5 \text{ m}}{(2 \text{ sec})^2} = \frac{50}{4} \frac{\text{kgm}}{(\text{sec})^2}$$

$$1 N' = 12.5 \text{ Newton} = 12.5 N$$

(ii) Value of 1 unit of energy in hypothetical

$$\text{system} = 1 \text{ J}' = \frac{10 \text{ kg} \times (5 \text{ m})^2}{(2 \text{ sec})^2} = 62.5 \frac{\text{kgm}^2}{\text{sec}^2}$$

$$= 62.5 \text{ Joule}$$

(iii) $1 \text{ J}' = 62.5 \text{ J}$

$$\Rightarrow 1 \text{ J} = \frac{1}{62.5} \text{ J}'$$

$$\Rightarrow 500 \text{ J} = 500 \times \frac{\text{J}'}{62.5} = 8 \text{ J}'$$

Ans. 8

Method-2:

$$n_1 u_1 = n_2 u_2$$

$$n_1 M_1 L_1^2 T_1^{-2} = n_2 M_2 L_2^2 T_2^{-2}$$

$$\Rightarrow 500 \times \frac{1 \text{ kg} \times 1 \text{ m}^2}{1 \text{ sec}^2} = n_2 \frac{(10 \text{ kg})(5 \text{ m})^2}{(2 \text{ sec})^2}$$

$$\Rightarrow 500 = n_2 \times \frac{250}{4}$$

$$n_2 = 4$$

Q. In a typical combustion engine the work done

by a gas molecule is given by $W = \alpha^2 \beta e^{\frac{\beta x^2}{kT}}$, where x is the displacement, k is the Boltzmann constant and T is the temperature. If α and β are constants, dimensions of α will be:

Sol. We have, $W = \alpha^2 \beta e^{\frac{\beta x^2}{kT}}$

$$\therefore [\beta x^2] = [kT]$$

$$= [ML^2 T^{-2}] \left[\because E = \frac{3}{2} kT \right]$$

$$\Rightarrow [\beta] = [MT^{-2}]$$

$$\text{And, } [W] = [\alpha^2 \beta] \Rightarrow [ML^2 T^{-2}] = [\alpha^2][MT^{-2}]$$

$$\Rightarrow [\alpha] = [L]$$

Q. Let us consider a system of units in which mass and angular momentum are dimensionless. If length has dimension of L , which of the following statement(s) is/are correct?

Sol. Mass = $M^0 L^0 T^0$

$$Mvr = M^0 L^0 T^0$$

$$M^0 \frac{L^1}{T^1} \cdot L^1 = M^0 L^0 T^0$$

$$L^2 = T^1 \quad \dots(i)$$

$$\text{Force} = M^1 L^1 T^{-2} \quad (\text{in SI})$$

$$= M^0 L^1 L^{-4} \quad [\text{In new system from equation (i)}]$$

$$= L^{-3}$$

$$\text{Energy} = M^1 L^2 T^{-2} \quad (\text{In SI})$$

$$= M^0 L^2 L^{-4} \quad [\text{In new system from equation (i)}]$$

$$= L^{-2}$$

$$\text{Power} = \frac{\text{Energy}}{\text{Time}}$$

$$= M^1 L^2 T^{-3} \quad (\text{in SI})$$

$$= M^0 L^2 L^{-6} \quad [\text{In new system from equation (i)}]$$

$$= L^{-4}$$

$$\text{Linear momentum} = M^1 L^1 T^{-1} \quad (\text{in SI})$$

$$= M^0 L^1 L^{-2} \quad [\text{In new system from equation (i)}]$$

$$= L^{-1}$$